

Programme of the Seminar of the Faculty of Mathematics and Physics, Charles University

Monday 18 May 2026

- Morning *Arrival, accommodation*
- 12:00 *Lunch*
- 14:30 – 15:30 Synergizing 3D Bioprinting, Stem Cells, and Tailored Bioreactor Systems to Advance Clinically Relevant Tissue Replacements (Assoc. Prof. Ing. Roman Matějka, Ph.D., Faculty of Biomedical Engineering, Czech Technical University in Prague)
- 15:30 – 16:00 *Break*
- 16:00 – 17:00 Advances in plasmonic biosensors and their applications in medicine (Mgr. Tomáš Špringer, Ph.D., Institute of Photonics and Electronics of the Czech Academy of Sciences)
- 17:00 – 18:00 Ultrafast Science at ELI: From Molecules to Materials (Ing. Shirly Josefina Espinoza Herrera, Ph.D., ELI Beamlines Facility)
- 18:00 – 18:20 Multi-exciton energy transfer, annihilation and correlation (Matteo Bruschi, Ph.D., Faculty of Mathematics and Physics, Charles University)
- 18:30 *Dinner*

Tuesday 19 May 2026

- 8:00 – 9:00 *Breakfast*
- 9:10 – 9:30 Ultrafast spectroscopy and the study of photosynthesis: a primer (Erika M. G. Keil, Ph.D., Faculty of Mathematics and Physics, Charles University)
- 9:30 – 10:30 Fingerprints of Strong Interactions between Two-Dimensional Materials and Metals (Ing. Matěj Velický, Ph.D., J. Heyrovský Institute of Physical Chemistry of the Czech Academy of Sciences)
- 10:30 – 11:00 *Coffee break*
- 11:00 – 12:00 Physical Chemistry of Ganglioside Nanodomains (Prof. Dr. rer. nat. Martin Peter Hof, DSc., J. Heyrovský Institute of Physical Chemistry of the Czech Academy of Sciences)
- 12:00 *Lunch*
- 14:30 – 15:30 Cold Plasma: From Lightning to Medicine and Clean Water (Ing. Petr Lukeš, Ph.D., Institute of Plasma Physics of the Czech Academy of Sciences)
- 15:30 – 16:00 *Break*

- 16:00 – 17:00 Terahertz plasmonics in graphene nanostructures (Assoc. Prof. RNDr. Petr Kužel, Ph.D., Institute of Physics of the Czech Academy of Sciences)
- 17:00 – 18:00 From MFF UK to Monoclonal Antibody Development at DIANA Biotechnologies (RNDr. Vojtěch Vyklický, Ph.D., DIANA Biotechnologies)
- 18:00 – 18:20 Synthesis and Characterization of Perovskite Single Crystals for Detector Applications (Ramashanker Gupta, Ph.D., Faculty of Mathematics and Physics, Charles University)
- 18:30 *Dinner*

Wednesday 20 May 2026

- 8:00 – 9:00 *Breakfast*
- 9:10 – 9:30 (How) does carbon offsetting work? (RNDr. Kateřina Charvátová, Faculty of Mathematics and Physics, Charles University)
- 9:30 – 10:30 Quantitative label-free single-molecule imaging (Ing. Barbora Špačková, Ph.D., Institute of Physics of the Czech Academy of Sciences)
- 10:30 – 11:00 *Coffee break*
- 11:00 – 12:00 Microscopy serving biologists – inspiration from bio-imaging core facility (Mgr. Aleš Benda, Ph.D., BIOCEV, Faculty of Science, Charles University)
- 12:00 *Lunch*
- 13:00 – 18:30 *Free afternoon – excursion*
- 18:30 *Dinner*

Thursday 21 May 2026

- 8:00 – 9:00 *Breakfast*
- 9:10 – 9:30 Epitaxial graphene electrodes for SiC detectors (Mgr. Bohdan Morzhuk, Faculty of Mathematics and Physics, Charles University)
- 9:30 – 10:30 From equation to product. How physics is turning into a market advantage (RNDr. Ing. Jan Kubát, Ph.D., Crytur)
- 10:30 – 11:00 *Coffee break*
- 11:00 – 12:00 Lab-Grown Startup: The Commercialization Journey of LAM-X Nanofiber Technology (RNDr. Roman Chaloupka, Ph.D., LAM-X)
- 12:00 *Lunch*
- 14:30 – 15:30 Interaction of intense electric field with proteins (Ing. Michal Cifra, Ph.D., Institute of Photonics and Electronics of the Czech Academy of Sciences)
- 15:30 – 16:00 *Break*

- 16:00 – 17:00 Cancer Isn't Just Mutations – It's Physics: Mechanosensing in Tumor Growth and Immunotherapy (RNDr. Adam Obr, Ph.D., Institute of Hematology and Blood Transfusion)
- 17:00 – 18:00 Beating More Information out of Thin Films Using Fs Pulses (Assoc. Prof. RNDr. Karel Židek, Ph.D., Institute of Plasma Physics of the Czech Academy of Sciences, TOPTEC Research Centre for Special Optics and Optoelectronic Systems)
- 18:00 – 18:20 Strain Relaxation Effects in Trilayer Epitaxial Graphene (Maria Baeva, Ph.D., Faculty of Mathematics and Physics, Charles University)
- 18:30 *Dinner*

Friday 22 May 2026

- 8:00 – 9:00 *Breakfast*
- 9:30 – 10:30 Clinical trials: When it (finally) starts to make sense (RNDr. Martin Kříž, Ph.D., Pharmnet)
- 10:30 – 11:00 *Coffee break*
- 11:00 – 12:00 Microdomains of biological membranes in the regulation of cellular metabolism (From structure to function) (Prof. RNDr. Jan Malinský, Ph.D., Institute of Experimental Medicine of the Czech Academy of Sciences)
- 12:00 *Lunch*
- 13:00 *Return to Prague*